$$e^{At} \triangleq \sum_{K=0}^{\infty} \frac{t^K}{K!} A^K$$

Can be computed via Laplace as:

$$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

$$= \mathcal{L}^{-1}\left\{ \frac{Q(s)}{r(s)} \right\}$$

where $Q(s)$ is $n \times n$ adjoint of $(sI - A)$ whose entries are polynomials in s of order $\leq n-1$

and $r(s) = \text{Det}(sI - A)$ an nth order poly in s

$\Rightarrow$ all entries of e^{At} are linear combinations of the scalar exp. fns $e^{\lambda_K t}$,

where λ_K eigenvalues of A (assumed distinct for now)

$=$

Another way to see this:
(less useful computationally, but quite helpful in certain theoretical arguments).

Recall Caley-Hamilton: for every $K \geq n$, A^K can be written as linear combination of $A^0 = I, A, A^2, \ldots, A^{n-1}$

This applies to each term in ∞ series for e^{At}, so that

$$\underbrace{\sum_{K=0}^{\infty} \frac{t^K}{K!} A^K}_{\infty \text{ series}} = \underbrace{\sum_{K=0}^{n-1} \alpha_K(t) A^K}_{\text{finite series (n terms)}}$$

The $\alpha_K(t)$ can be found from

$$\underbrace{\begin{bmatrix} 1 & \lambda_1 & \lambda_1^2 & \cdots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \lambda_2^2 & & \lambda_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \lambda_n & \lambda_n^2 & & \lambda_n^{n-1} \end{bmatrix}}_{\text{"Vandermonde matrix"}} \begin{bmatrix} \alpha_1(t) \\ \alpha_2(t) \\ \vdots \\ \alpha_n(t) \end{bmatrix} = \begin{bmatrix} e^{\lambda_1 t} \\ e^{\lambda_2 t} \\ \vdots \\ e^{\lambda_n t} \end{bmatrix}$$

$\Rightarrow$ each $\alpha_K(t)$ is a linear comb. of $e^{\lambda_K t}$ terms, λ_K eigenvalue of A.

Is there significance also of the eigenvectors of A in "unpacking" e^{At}?

recall, for each λ_k, eigenvector $\underline{v}_k$ satisfies

$$A\underline{v}_k = \lambda_k \underline{v}_k, \quad \underline{v}_k \neq \underline{0}$$

$$\Rightarrow \underline{v}_k \in \mathcal{N}(\lambda_k I - A)$$

$$= \det V = [\underline{v}_1 \underline{v}_2 \cdots \underline{v}_n] \quad \text{(eigenvector matrix)}$$

Then

$$AV = [(A\underline{v}_1)\ (A\underline{v}_2) \cdots (A\underline{v}_n)]$$

$$= [(\lambda_1 \underline{v}_1)\ (\lambda_2 \underline{v}_2) \cdots (\lambda_n \underline{v}_n)]$$

$$= V\Lambda$$

where $\Lambda = \text{diag}\{\lambda_k\}_{k=1}^{n}$

(Diagonal matrix of eigenvalues)

so $\boxed{AV = V\Lambda}$

$A = V\Lambda V^{-1}$ $\quad$ $\Lambda = V^{-1}AV$

(assuming V^{-1} exists!)

Fact: The eigenvectors for distinct eigenvalues are linearly indep.

$\Rightarrow$ cols of V lin. indep

$\Rightarrow \det V \neq 0$

$\Rightarrow V^{-1}$ exists.

$\simeq$

How to exploit?

Note that, for any K

$$A^K = (V\Lambda V^{-1})^K$$

$$= (V\Lambda V^{-1})(V\Lambda V^{-1})(V\Lambda V^{-1}) \cdots (V\Lambda V^{-1})$$

K TIMES

so $A^K = V\Lambda^K V^{-1}$

when $K = 0$, $A^0 = I = VV^{-1}$

Thus

$$e^{At} = \sum_{K=0}^{\infty} \frac{t^K}{K!} A^K$$

$$\boxed{e^{At} = V\left[\sum_{K=0}^{\infty} \frac{t^K}{K!}\Lambda^K\right]V^{-1}}$$

$$e^{\Lambda t} = \text{diag}\{e^{\lambda_k t}\}$$

Summary (so far)

$$e^{At} = V e^{\Lambda t} V^{-1}$$

V - eigenvector matrix for A

$e^{\Lambda t} = \text{diag}\{e^{\lambda_k t}\}$ λ_k eigenvalue

Now let

$$V^{-1} = \begin{bmatrix} \underline{\omega}_1^T \\ \underline{\omega}_2^T \\ \vdots \\ \underline{\omega}_n^T \end{bmatrix}$$ (row partition)

i.e. $\underline{\omega}_k^T = k^{th}$ row of V^{-1}

So that

$$e^{At} = [\underline{v}_1\ \underline{v}_2 \cdots \underline{v}_n] \begin{bmatrix} e^{\lambda_1 t} & & & \\ & e^{\lambda_2 t} & & \\ & & \ddots & \\ & & & e^{\lambda_n t} \end{bmatrix} \begin{bmatrix} \underline{\omega}_1^T \\ \underline{\omega}_2^T \\ \vdots \\ \underline{\omega}_n^T \end{bmatrix}$$

$$= [\underline{v}_1\ \underline{v}_2 \cdots \underline{v}_n] \begin{bmatrix} e^{\lambda_1 t}\underline{\omega}_1 \\ e^{\lambda_2 t}\underline{\omega}_2 \\ \vdots \\ e^{\lambda_n t}\underline{\omega}_n \end{bmatrix}$$

$$= e^{\lambda_1 t}\underline{v}_1\underline{\omega}_1^T + \cdots + e^{\lambda_n t}\underline{v}_n\underline{\omega}_n^T$$

Thus

$$e^{At} = \sum_{k=1}^{n} e^{\lambda_k t}\underline{v}_k\underline{\omega}_k^T$$

"Dyadic expansion" of e^{At}

"dyad" (expands to nxn matrix after mult)

What are these $\underline{\omega}_k^T$ terms (beyond the purely technical def'n above?)

Consider

$$A V = V \Lambda$$

$$\Rightarrow \quad V^{-1} A = \Lambda V^{-1}$$

$$\begin{bmatrix} \underline{\omega}_1^T \\ \underline{\omega}_2^T \\ \vdots \\ \underline{\omega}_n^T \end{bmatrix} A = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} \underline{\omega}_1^T \\ \underline{\omega}_2^T \\ \vdots \\ \underline{\omega}_n^T \end{bmatrix}$$

$$\begin{bmatrix} \underline{\omega}_1^T A \\ \underline{\omega}_2^T A \\ \vdots \\ \underline{\omega}_n^T A \end{bmatrix} = \begin{bmatrix} \lambda_1 \underline{\omega}_1^T \\ \lambda_2 \underline{\omega}_2^T \\ \vdots \\ \lambda_n \underline{\omega}_n^T \end{bmatrix}$$

$$\Rightarrow \quad \underline{\omega}_K^T A = \lambda_K \underline{\omega}_K^T$$

Compare with

$$A \underline{v}_K = \lambda_K \underline{v}_K$$

$\underline{v}_K$ "Right" eigenvectors (columns)
$\underline{\omega}_K^T$ "Left" eigenvectors (rows)

Equivalently

$$\left(\underline{\omega}_K^T A\right)^T = \left(\lambda_K \underline{\omega}_K^T\right)^T$$

$$A^T \underline{\omega}_K = \lambda_K \underline{\omega}_K$$

$\Rightarrow$ $\underline{\omega}_K$ are the ordinary (right) eigenvectors of A^T.

Note: A and A^T have same char poly, hence same eigenvalues λ_K.

$$\text{Det}\, A^T = \text{Det}\, A$$

so $\text{Det}(sI - A)^T = \text{Det}(sI - A)$

$$\text{Det}(sI - A^T) = \text{Det}(sI - A)$$

Now, $V^{-1} V = I$

or $\begin{bmatrix} \underline{\omega}_1^T \\ \vdots \\ \underline{\omega}_n^T \end{bmatrix} [\underline{v}_1 \cdots \underline{v}_n] = I$

So that

$$\begin{bmatrix} \underline{\omega}_1^T\underline{v}_1 & \underline{\omega}_1^T\underline{v}_2 & \cdots & \underline{\omega}_1^T\underline{v}_n \\ \underline{\omega}_2^T\underline{v}_1 & \cdots & & \underline{\omega}_2^T\underline{v}_n \\ \vdots & & & \\ \underline{\omega}_n^T\underline{v}_1 & \cdots & & \underline{\omega}_n^T\underline{v}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 \cdots & 0 \\ & & & \\ & & & \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Hence

$$\Rightarrow \underline{\omega}_i^T\underline{v}_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$$

Thus $\{\underline{v}_k\}_{k=1}^n$ and $\{\underline{\omega}_k\}_{k=1}^n$ are "bi-orthonormal" sets.

Example application: calc. e^{At} when

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$$

$\lambda_1 = -2$ $\quad$ $\lambda_2 = -3$

$$\underline{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad \underline{v}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix} \quad V = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

Calculate

$$V^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{matrix} \leftarrow \underline{\omega}_1^T \\ \leftarrow \underline{\omega}_2^T \end{matrix}$$

$$\underline{v}_1\underline{\omega}_1^T = \begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} \quad \underline{v}_2\underline{\omega}_2^T = \begin{bmatrix} -2 & -1 \\ 6 & 3 \end{bmatrix}$$

$$e^{At} = e^{-2t}\begin{bmatrix} 3 & 1 \\ -6 & -2 \end{bmatrix} + e^{-3t}\begin{bmatrix} -2 & -1 \\ 6 & 3 \end{bmatrix}$$

Example appl #2: Solving homog eq'n

$$\dot{\underline{x}}(t) = A\underline{x}(t), \quad \underline{x}(0) = \underline{x}_0$$

We know $\underline{x}(t) = e^{At}\underline{x}_0$

Use dyadic expansion

Dot products!

$$\underline{x}(t) = \left[\sum_{k=1}^{n} e^{\lambda_k t}\, \underline{v}_k \underline{\omega}_k^T\right]\underline{x}_0$$

or

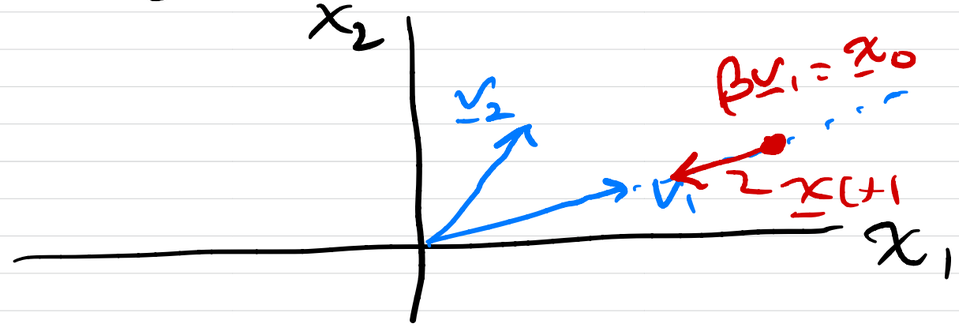
$$\underline{x}(t) = \sum_{k=1}^{n} (\underline{\omega}_k^T \underline{x}_0)\, e^{\lambda_k t}\, \underline{v}_k$$

- $(\underline{\omega}_k^T \underline{x}_0)$: amount of kth mode present in response (constant) = extent k^{th} mode is "excited" by ICs
- $e^{\lambda_k t}\underline{v}_k$: "modes" of state eq'n

Suppose, for example

$\underline{x}_0 = \beta \underline{v}_j$ for some $j = 1, \ldots, n$

(i.e. IC lies along one of the right eigenvectors)



Then

$$\underline{x}(t) = \sum_{k=1}^{n} \beta(\underline{\omega}_k^T \underline{v}_j)\, e^{\lambda_k t}\, \underline{v}_k$$

$(\underline{\omega}_k^T \underline{v}_j) = 0 \quad k \neq j$

$= 1 \quad k = j$

$$\Rightarrow \underline{x}(t) = \beta e^{\lambda_j t}\underline{v}_j \text{ for all } t$$

Only j^{th} mode present in response!

(No other modes contribute to $\underline{x}(t)$ for such an IC)